Assignment: Fourier Series Representation of an Animal Contour

Objective

The objective of this assignment is to apply the concepts of **complex Fourier series** to the reconstruction of a two-dimensional contour. Based on the approach presented in the video *"But what is a Fourier series? From heat flow to drawing with circles"* by 3Blue1Brown, students will represent the contour of an animal as a parametric complex-valued function and determine the first 100 Fourier-series coefficients required to approximate the selected contour.

In the referenced video, a planar drawing is interpreted as a complex-valued function whose Fourier representation can be understood as a sum of rotating vectors. The coefficients determine the magnitude and initial phase of these vectors, while their integer frequencies determine their angular velocities. As more Fourier terms are included, the reconstructed curve approaches the original drawing.

Instructions

1. **Select an animal.**

   Choose your favorite pet or any other animal of your preference. The selected animal should have a clearly identifiable two-dimensional contour.

2. **Obtain a contour.**

   Obtain or create a suitable image of the selected animal and extract a single closed contour representing its external boundary. The contour should be represented by a sequence of ordered points

   $$(x_k, y_k), \qquad k = 0, \ldots, N-1,$$

   distributed along the contour.

3. **Construct the complex-valued signal.**

   Represent the contour using the complex variable

   $$z(t) = x(t) + iy(t), \qquad 0 \leq t < 1,$$

   where $x(t)$ and $y(t)$ describe the horizontal and vertical coordinates of the contour, respectively.

4. **Compute the Fourier coefficients.**

   Using the complex Fourier-series formulation,

   $$z(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi jkt},$$

compute the coefficients

$$c_n = \int_0^1 z(t) e^{-2\pi j k t}\, dt.$$

For a numerically sampled contour, the coefficients may be approximated using numerical integration or the discrete Fourier transform (DFT), provided that the relationship between the numerical approximation and the continuous Fourier-series formulation is clearly explained.

5. **Use the first 100 terms.**

   Determine the **100 Fourier coefficients with the frequencies selected for the reconstruction**. Clearly state how the 100 terms are distributed between positive and negative frequencies. For example, a symmetric selection may be written as

   $$n = -50, -49, \ldots, -1, 0, 1, \ldots, 49,$$

   giving 100 coefficients.

   If a different convention is used, it must be explicitly justified.

6. **Reconstruct the contour.**

   Using the selected Fourier coefficients, reconstruct the contour according to

   $$z_{100}(t) = \sum_{n \in \mathcal{N}_{100}} c_n e^{2\pi j k t},$$

   where $\mathcal{N}_{100}$ denotes the set of 100 selected frequencies.

7. **Visualize the result.**

   Plot both:

   - the original animal contour; and
   - the reconstructed contour obtained from the 100 Fourier terms.

   The two curves should be displayed on the same coordinate system to facilitate comparison.

8. **Interpret the results.**

   Discuss the quality of the reconstruction. In particular, comment on:

   - which geometric features of the animal are accurately reproduced;
   - which details are lost or distorted;
   - the effect of increasing or decreasing the number of Fourier terms;
   - the relationship between the Fourier coefficients and the rotating-vector interpretation presented in the reference video.

Computational Requirements

The analysis must be implemented using a computational tools in **Julia**.

The submitted solution should include sufficient computational details to allow another student to reproduce the results.

At minimum, the following should be reported:

- the selected animal;
- the source or method used to obtain the contour;

- the number of sampled contour points;
- the parameterization of the contour;
- the Fourier coefficients;
- the selected frequency indices;
- the reconstructed contour;
- an appropriate visualization comparing the original and reconstructed contours. **It is highly recommended to produce an animation of the reconstruction.**

Report Requirements

The final report must be written in **LaTeX** and submitted as a **PDF**. The report should contain, at minimum, the following sections:

1. Introduction
2. Mathematical Background
3. Contour Acquisition and Parameterization
4. Fourier-Series Computation
5. Reconstruction Using 100 Terms
6. Results and Discussion
7. Conclusions
8. References

Group Work

The assignment must be completed in **groups of up to three students**. Each group is expected to collaborate on the mathematical development, computational implementation, and analysis. However, **each student must submit the solution individually**. Each submission must therefore contain the student's own report and clearly identify the members of the group with whom the work was carried out.

Important

The purpose of the assignment is not simply to obtain a visually attractive drawing. The report must demonstrate the mathematical connection between the selected contour, its complex Fourier representation, and the reconstruction obtained from the first 100 Fourier terms.

References

[1] 3Blue1Brown, *But what is a Fourier series? From heat flow to drawing with circles*, YouTube, 2019. https://www.youtube.com/watch?v=r6sGWTCMz2k